

HAAST, New Zealand

WMO: 937090

Lat: 43.8604S

Lon: 169.0070E

Elev: 5

StdP: 101.26

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.3	2.3	-7.6	2.0	8.0	-5.1	2.5	7.7	12.0	9.9	10.6	11.2	1.9	150	0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.9	20.6	17.2	19.7	16.7	19.0	16.3	18.4	19.6	17.7	18.9	17.1	18.4	3.7	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.9	12.9	19.0	17.2	12.3	18.4	16.5	11.7	17.9	52.2	19.7	50.0	19.0	48.0	18.4	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	9.6	8.2	-1.1	22.9	1.0	1.4	-1.8	23.9	-2.4	24.7	-3.0	25.5	-3.8	26.6		
			-1.7	19.2	1.1	0.9	-2.5	19.8	-3.1	20.3	-3.7	20.8	-4.5	21.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	14.5	14.9	13.8	12.3	10.6	8.5	7.9	8.4	9.3	10.1	11.4	13.2
	DBStd	3.09	2.07	1.94	2.01	1.94	2.24	2.02	1.98	1.90	1.80	1.88	1.87	2.09
	HDD10.0	262	0	0	1	4	18	53	69	55	34	21	7	1
	HDD18.3	2594	119	96	142	182	238	294	322	307	272	254	209	158
	CDD10.0	713	141	138	117	72	38	8	6	7	12	24	48	102
	CDD18.3	3	2	1	1	0	0	0	0	0	0	0	0	0
	CDH23.3	4	3	1	1	0	0	0	0	0	0	0	0	0
	CDH26.7	1	1	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	3.0	2.7	2.4	2.6	2.8	3.0	3.2	3.3	3.0	3.2	3.3	3.2	2.9
Precipitation	PrecAvg	3770	304	237	292	268	329	349	269	298	345	369	320	376
	PrecMax	4832	611	438	605	491	675	641	456	593	690	624	550	789
	PrecMin	3261	72	70	126	94	135	156	96	148	100	196	77	197
	PrecStd	412	123	99	115	119	106	127	92	104	135	108	123	135
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.6	21.7	20.7	18.6	17.3	14.7	14.3	15.2	15.8	18.1	19.0	20.7
		MCWB	17.5	17.6	16.1	14.1	13.0	11.4	9.1	9.3	10.2	11.1	14.8	17.3
	2%	DB	20.6	20.6	19.4	17.3	15.8	13.7	13.2	13.5	14.3	15.7	17.0	19.2
		MCWB	17.6	17.5	16.2	14.3	13.0	11.8	9.7	9.9	11.1	11.8	13.9	16.4
	5%	DB	19.6	19.7	18.6	16.6	15.0	12.9	12.3	12.6	13.6	14.6	16.1	18.2
		MCWB	16.9	17.0	15.9	14.0	13.0	10.9	9.5	10.0	11.0	11.9	13.5	15.7
	10%	DB	18.7	18.9	17.8	15.9	14.3	12.1	11.5	11.9	12.9	13.9	15.3	17.2
		MCWB	16.3	16.6	15.4	13.7	12.5	9.9	8.9	9.7	10.8	11.6	13.1	15.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.6	19.2	18.7	16.6	15.3	13.7	12.7	12.5	13.1	14.2	16.2	18.0
		MCDB	21.1	20.3	19.3	17.4	15.9	14.0	13.1	12.9	13.7	15.2	17.7	19.6
	2%	WB	18.3	18.4	17.7	15.7	14.4	12.7	11.7	11.6	12.5	13.4	15.0	17.1
		MCDB	19.7	19.6	18.5	16.4	15.0	13.3	12.4	12.4	13.3	14.3	16.0	18.5
	5%	WB	17.5	17.8	16.8	15.0	13.7	11.7	10.7	10.8	11.9	12.9	14.2	16.4
		MCDB	18.8	19.1	17.9	15.9	14.5	12.4	11.5	11.9	13.0	13.7	15.4	17.6
	10%	WB	16.8	17.1	16.0	14.2	12.8	10.6	9.6	10.1	11.4	12.2	13.6	15.6
		MCDB	18.2	18.5	17.3	15.4	13.9	11.5	10.7	11.5	12.5	13.4	14.8	16.8
Mean Daily Temperature Range	5% DB	MDBR	6.6	6.9	6.9	6.7	6.0	6.2	6.4	6.5	6.7	6.8	6.6	6.3
		MCDBR	7.8	7.7	7.4	7.0	6.0	6.4	6.7	6.9	7.5	7.6	8.0	7.3
		MCWBR	5.6	5.5	5.3	5.2	4.5	4.9	4.8	5.0	5.6	5.4	5.9	5.3
	5% WB	MCDBR	6.3	6.4	5.7	5.8	4.9	5.1	5.5	6.0	6.0	5.6	6.4	5.9
		MCWBR	4.9	5.0	4.8	5.1	4.5	4.8	4.9	5.0	5.0	4.9	5.1	4.6
Clear-Sky Solar Irradiance	taub	0.347	0.337	0.320	0.312	0.298	0.293	0.284	0.301	0.337	0.340	0.337	0.339	
	taud	2.535	2.582	2.631	2.633	2.643	2.601	2.627	2.610	2.489	2.493	2.526	2.558	
	Ebn at Noon	977	959	931	865	806	767	813	865	894	944	980	991	
	Edh at Noon	107	96	83	71	58	55	57	70	94	104	107	106	
All-Sky Solar Radiation	RadAvg	5.98	5.42	4.01	2.69	1.60	1.30	1.51	2.23	3.30	4.49	5.72	6.17	
	RadStd	0.45	0.45	0.26	0.26	0.13	0.13	0.13	0.18	0.33	0.38	0.55	0.41	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (2 neighbors)	N/A	N/A	+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page